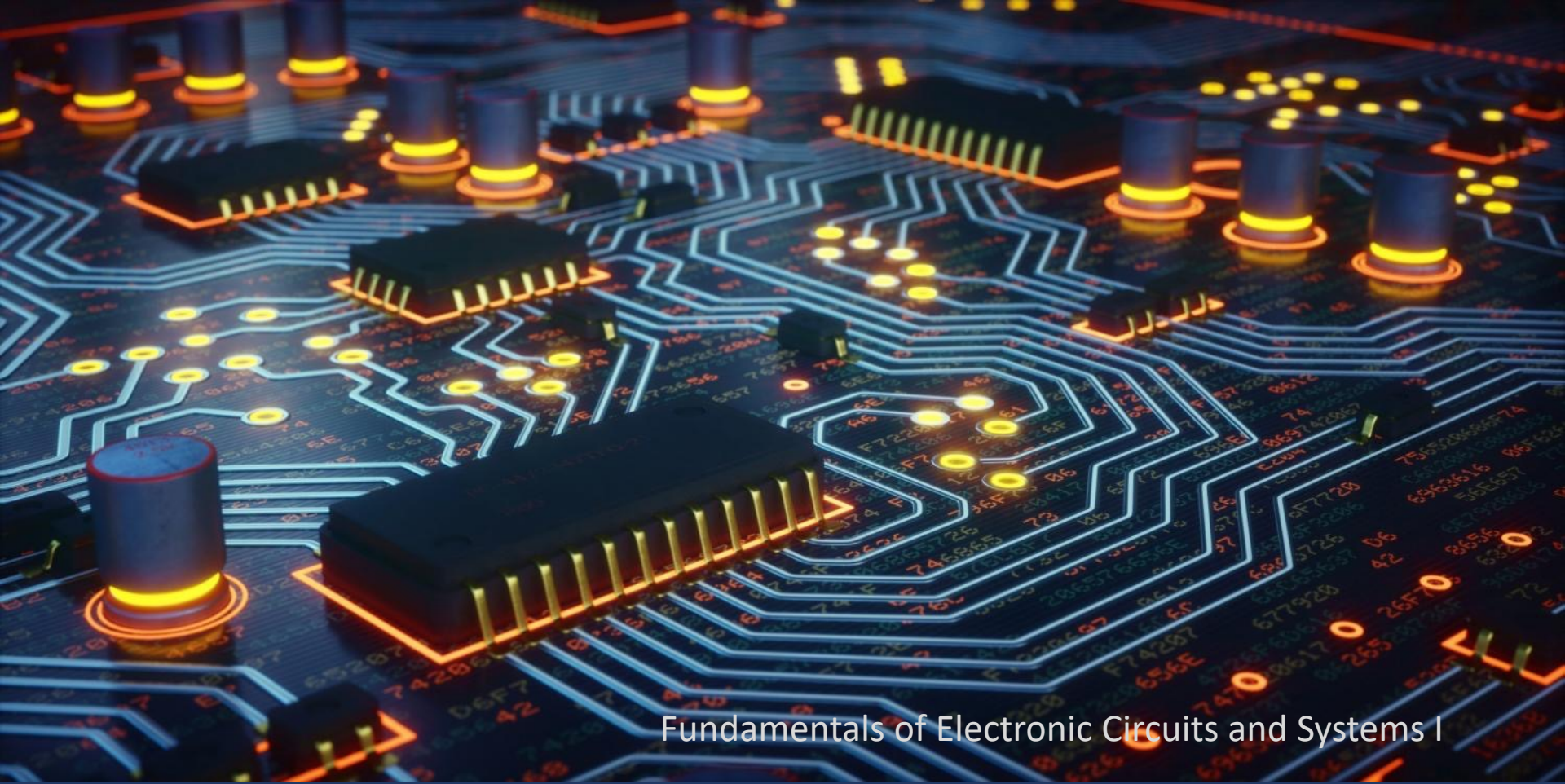




Fundamentals of Electronic Circuits and Systems I

AC Steady-State Analysis

Milin Zhang

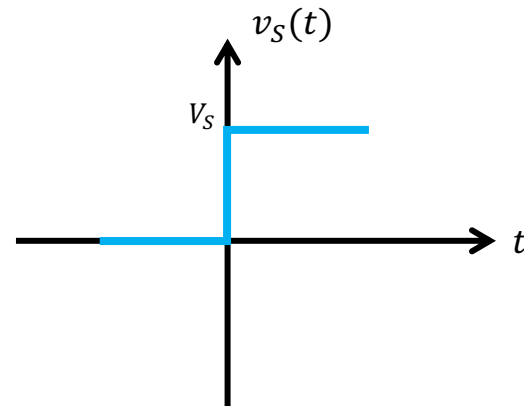
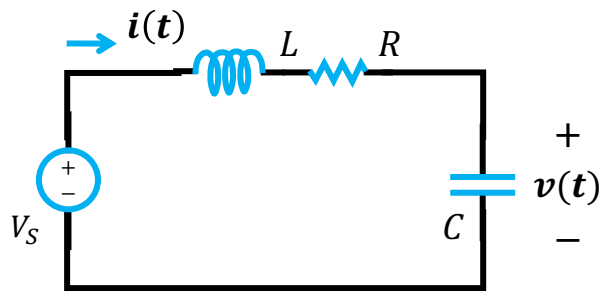
Dept of EE, Tsinghua University



Recall

- How to analyze 1st/2nd order circuit in time domain?
 - Write the circuit equation according to KVL/KCL
 - Solve the differential equation
 - Step 1a: Find the **particular integral solution** $v_p(t)$
 - Step 1b: substitute $v_p(t)$ to the equation to solve the unknown
 - Step 2a: find the **homogeneous equation**
 - Step 2b: find the **complementary solution** $v_c(t)$ to the homogeneous equation
 - Step 3a: find the initial voltage/current values
 - Step 3b: substitute the initials to the full solution to solve the unknown

Recall: Particular integral solution to step forcing func.



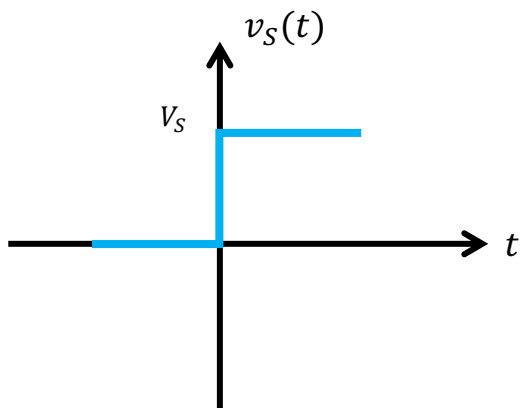
- Circuit equations according to KVL/KCL
$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = \frac{V_S}{LC}$$
- Particular integral solution
$$v_p(t) = V_S$$



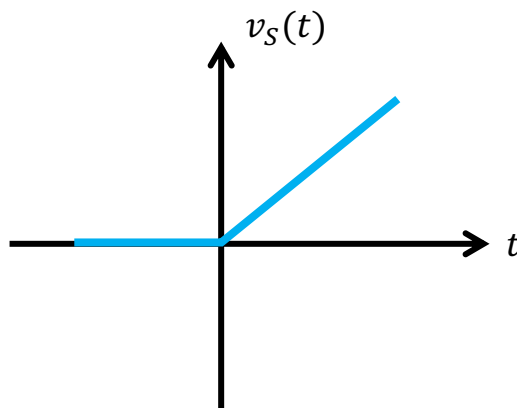
Outlines

- Response to different forcing function

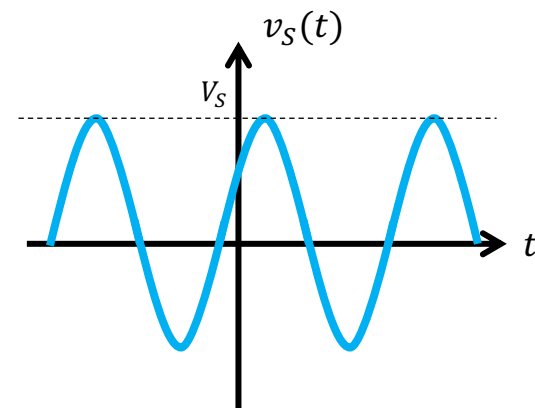
More forcing functions



Step function



Ramp function

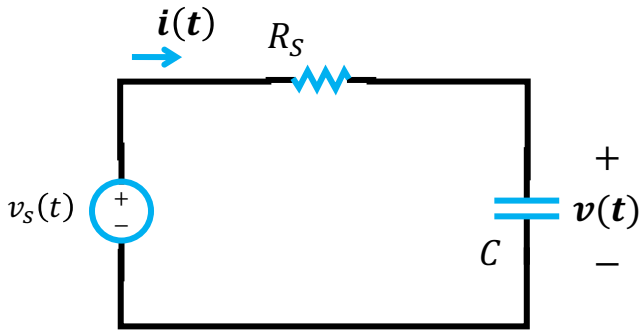


Sinusoidal function

Assume $x_p(t) = K$

Example 1: ramp forcing function

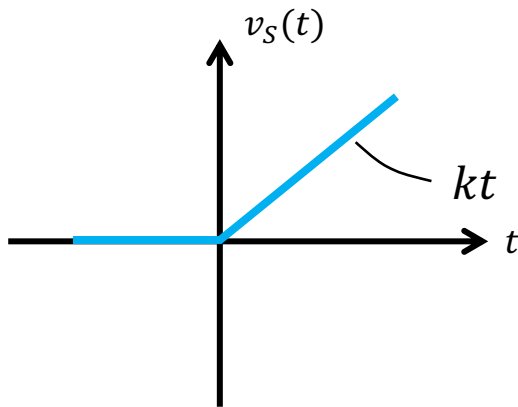
QUESTION: Assume there is no charge on the capacitor C at $t = -\infty$. Find the voltage response to the ramping forcing function.



- According to KVL

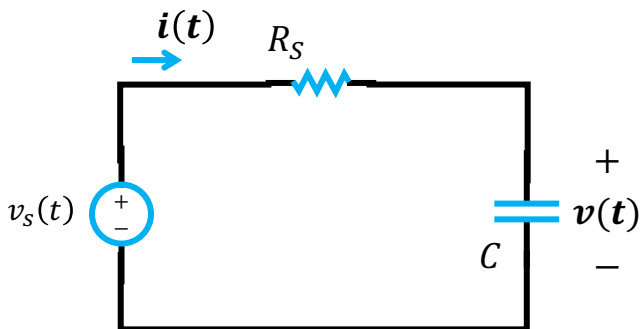
$$i_R R + v(t) = v_s(t)$$

$$\rightarrow \frac{d}{dt}v(t) + \frac{1}{RC}v(t) = \frac{1}{RC}kt$$



Example 1: ramp forcing function

QUESTION: Assume there is no charge on the capacitor C at $t = -\infty$. Find the voltage response to the ramping forcing function.



$$\frac{d}{dt}v(t) + \frac{1}{RC}v(t) = \frac{1}{RC}kt$$

- Step 1a: find the **particular integral solution** $v_p(t)$

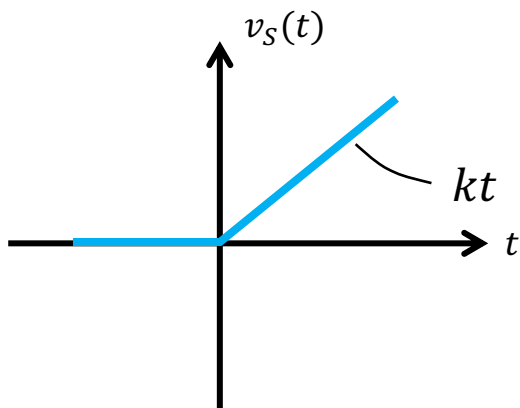
$$\text{Assume } v_p(t) = pt + q$$

- Step 1b: substitute $v_p(t)$ to the equation

$$p + \frac{pt + q}{RC} = \frac{1}{RC}kt$$

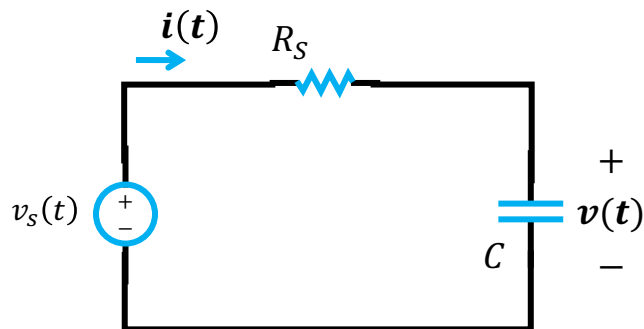
$$\rightarrow pt + (RCp + q) = kt$$

$$\rightarrow p = k, q = -RCk$$



Example 1: ramp forcing function

QUESTION: Assume there is no charge on the capacitor C at $t = -\infty$. Find the voltage response to the ramping forcing function.



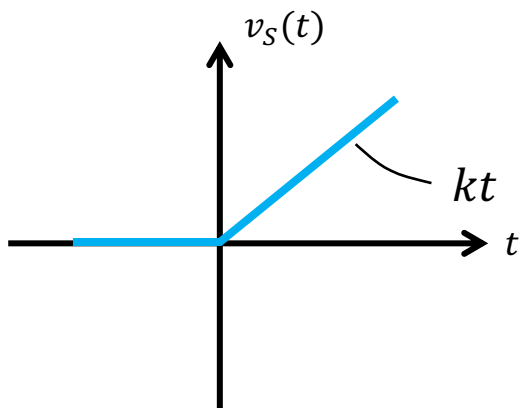
$$\frac{d}{dt}v(t) + \frac{1}{RC}v(t) = \frac{1}{RC}kt$$

- Step 1: find the **particular integral solution** $v_p(t)$

$$v_p(t) = kt - RCq$$

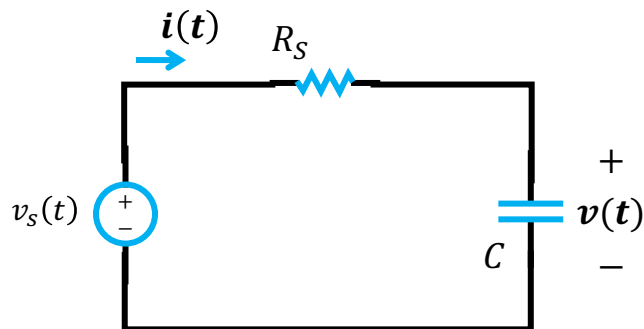
- Step 2: find the **complementary solution** $v_c(t)$

$$v_c(t) = K_2 e^{-a_1 t} \quad \text{where } a_1 = \frac{1}{RC}$$



Example 1: ramp forcing function

QUESTION: Assume there is no charge on the capacitor C at $t = -\infty$. Find the voltage response to the ramping forcing function.



$$\frac{d}{dt}v(t) + \frac{1}{RC}v(t) = \frac{1}{RC}kt$$

- Full solution

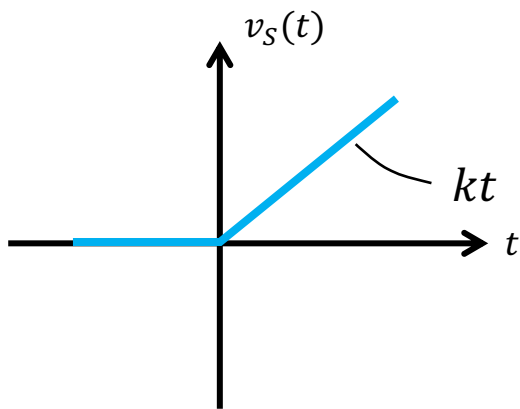
$$v(t) = kt - RCk + K_2e^{-a_1t} \quad \text{where } a_1 = \frac{1}{RC}$$

- Step 3a: find the value of $v(t)$ at one instant of time

$$v(0) = 0$$

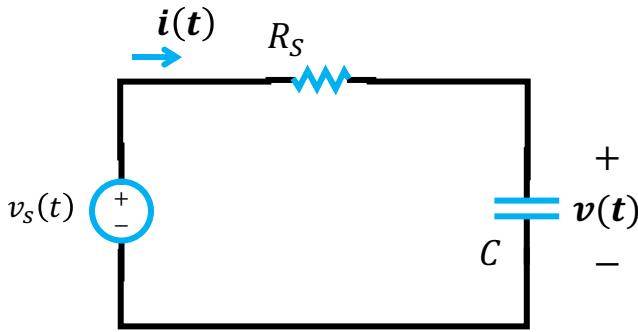
- Step 3b: substitute $v(0) = 0$ to $v(t)$

$$\rightarrow K_2 = RCk$$



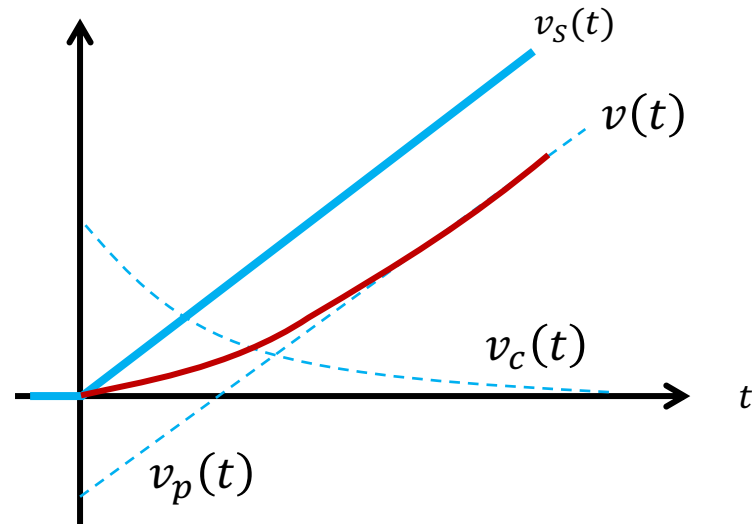
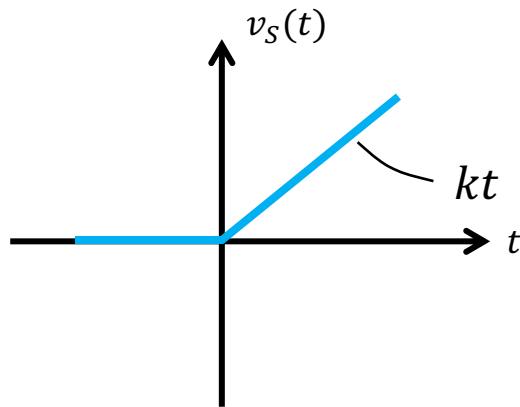
Example 1: ramp forcing function

QUESTION: Assume there is no charge on the capacitor C at $t = -\infty$. Find the voltage response to the ramping forcing function.



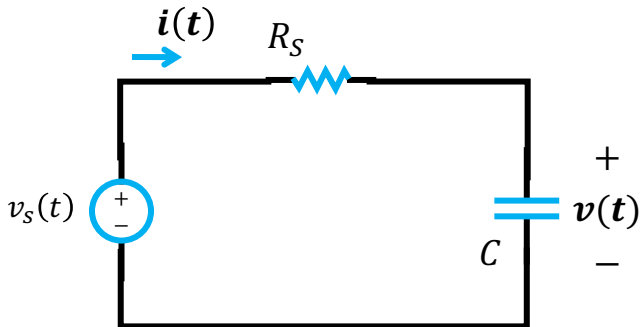
$$\frac{d}{dt}v(t) + \frac{1}{RC}v(t) = \frac{1}{RC}kt$$

Solution $v(t) = kt - RCk + RCke^{-\frac{1}{RC}}$

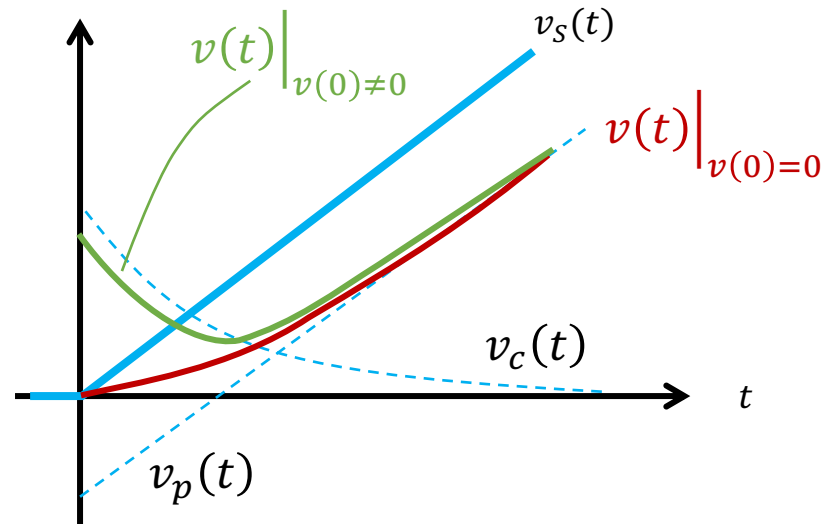
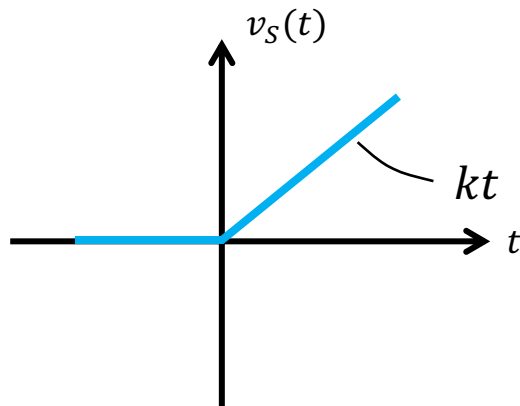


Example 1: ramp forcing function

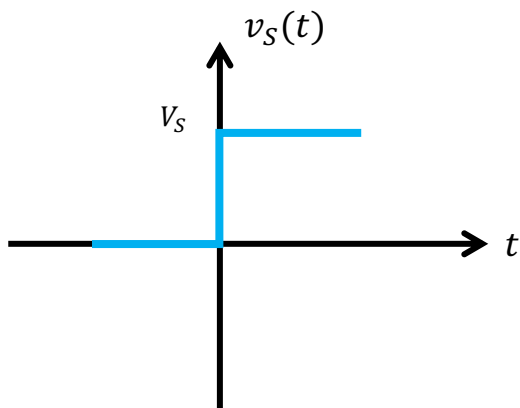
QUESTION: Assume there is no charge on the capacitor C at $t = -\infty$. Find the voltage response to the ramping forcing function.



What if $v(0) \neq 0$?

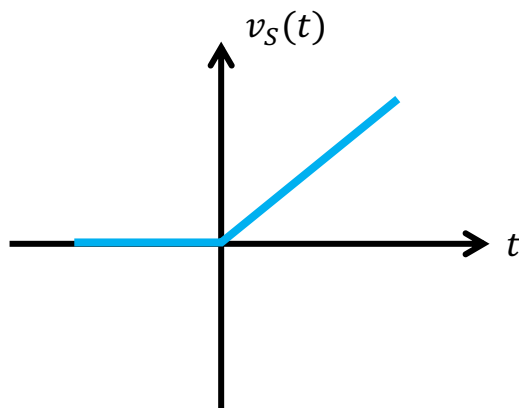


More forcing functions



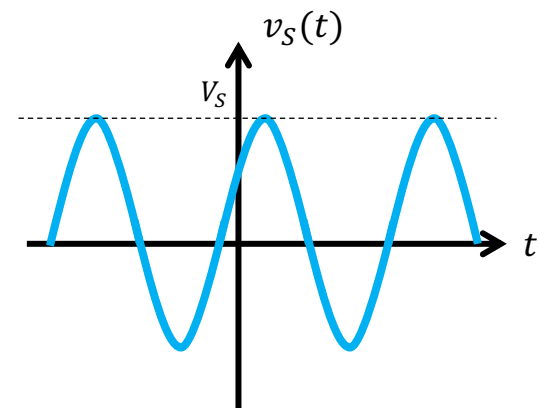
Step function

Assume $x_p(t) = K$



Ramp function

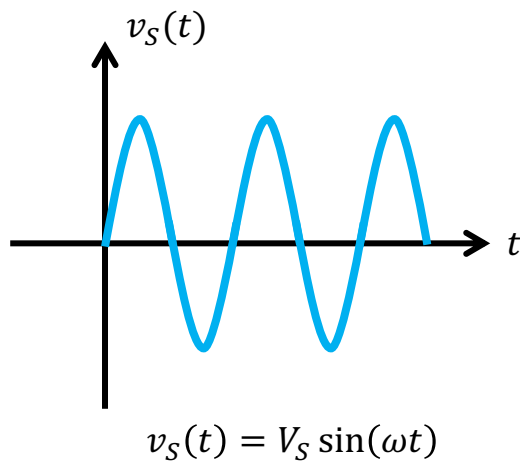
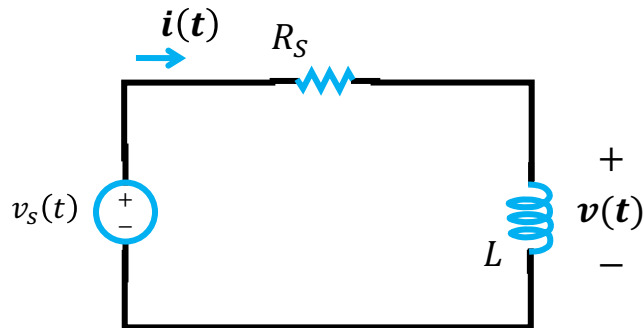
Assume $x_p(t) = pt + q$



Sinusoidal function

Example 2: Sinusoidal forcing function

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to the sinusoidal forcing function.



- According to KVL

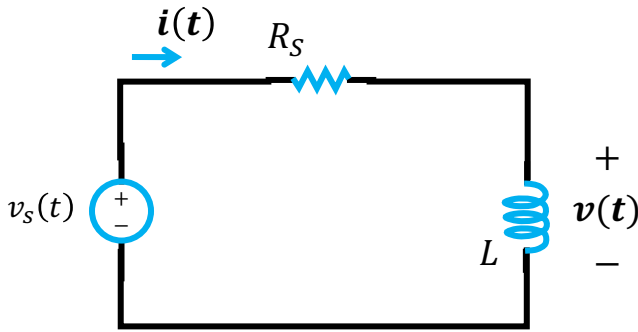
$$i(t)R + v_L(t) = v_S(t)$$

$$\rightarrow i(t)R + L \frac{d}{dt} i(t) = v_S(t)$$

$$\rightarrow \frac{d}{dt} i(t) + \frac{R}{L} i(t) = \frac{1}{L} v_S(t)$$

Example 2: Sinusoidal forcing function

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to the sinusoidal forcing function.



$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = \frac{1}{L}v_s(t)$$

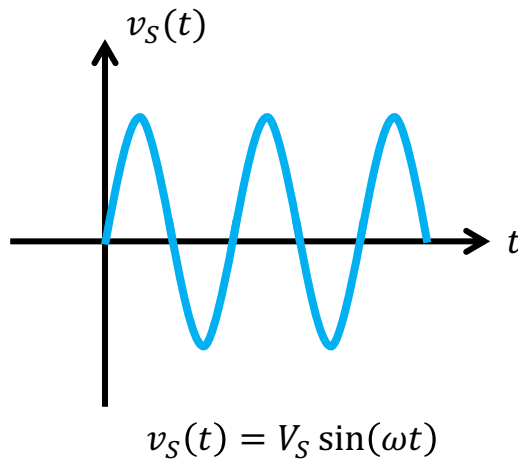
- Step 1a: find the **particular integral solution $i_p(t)$**

$$\text{Assume } i_p(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t)$$

- Step 1b: substitute $i_p(t)$ to the equation

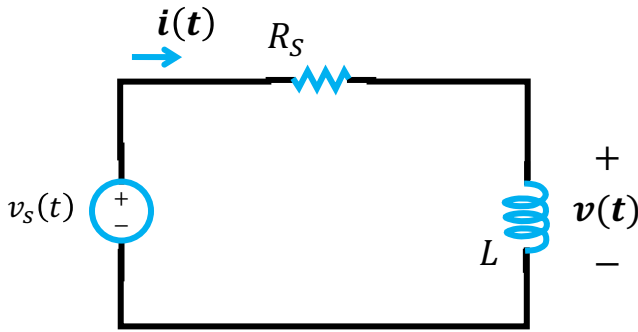
$$V_S \sin(\omega t) = K_1 R \cos(\omega t) + K_2 R \sin(\omega t) + L\omega K_2 \cos(\omega t) - L\omega K_1 \sin(\omega t)$$

$$\rightarrow \begin{cases} V_S = K_2 R - L\omega K_1 \\ 0 = K_1 R + L\omega K_2 \end{cases} \rightarrow \begin{cases} K_1 = V_S \frac{-\omega L}{R^2 + \omega^2 L^2} \\ K_2 = V_S \frac{R}{R^2 + \omega^2 L^2} \end{cases}$$



Example 2: Sinusoidal forcing function

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to the sinusoidal forcing function.



$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = \frac{1}{L}v_s(t)$$

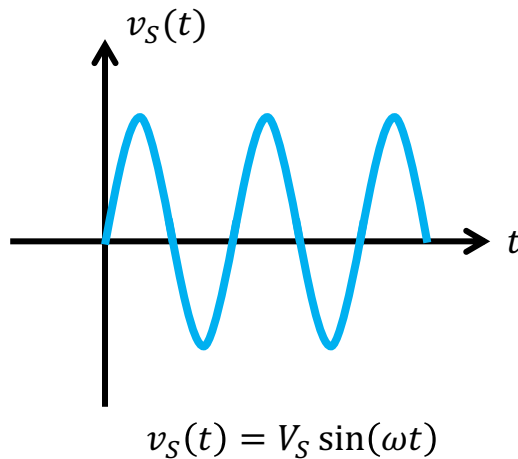
- Step 1: find the **particular integral solution** $i_p(t)$

$$i_p(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t)$$

$$\text{where } \begin{cases} K_1 = V_S \frac{-\omega L}{R^2 + \omega^2 L^2} \\ K_2 = V_S \frac{R}{R^2 + \omega^2 L^2} \end{cases}$$

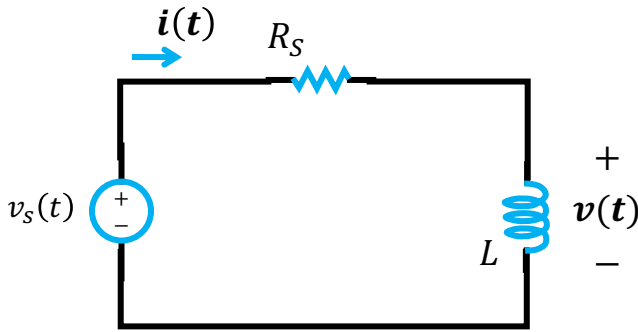
- Step 2: find the **complementary solution** $i_c(t)$

$$i_c(t) = K_3 e^{-\frac{R}{L}t}$$



Example 2: Sinusoidal forcing function

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to the sinusoidal forcing function.



$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = \frac{1}{L}v_s(t)$$

- Full solution

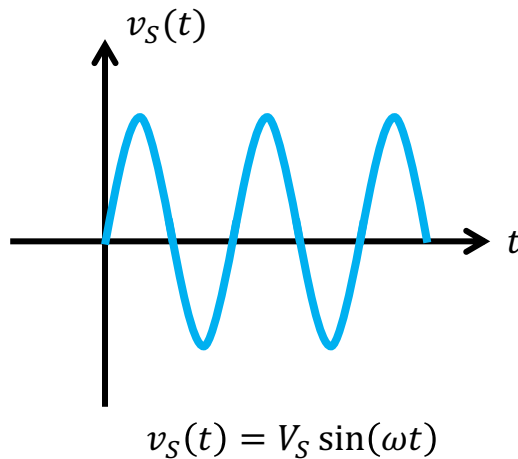
$$i(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t) + K_3 e^{-\frac{R}{L}t}$$

- Step 3a: find the value of $i(t)$ at one instant of time

$$i(0) = 0$$

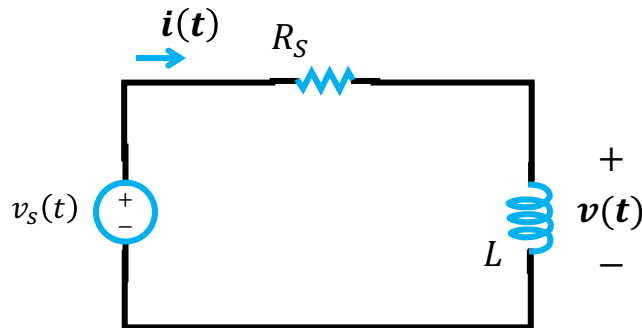
- Step 3b: substitute $i(0) = 0$ to $i(t)$

$$\rightarrow K_3 = \frac{\omega L V_S}{R^2 + \omega^2 L^2}$$



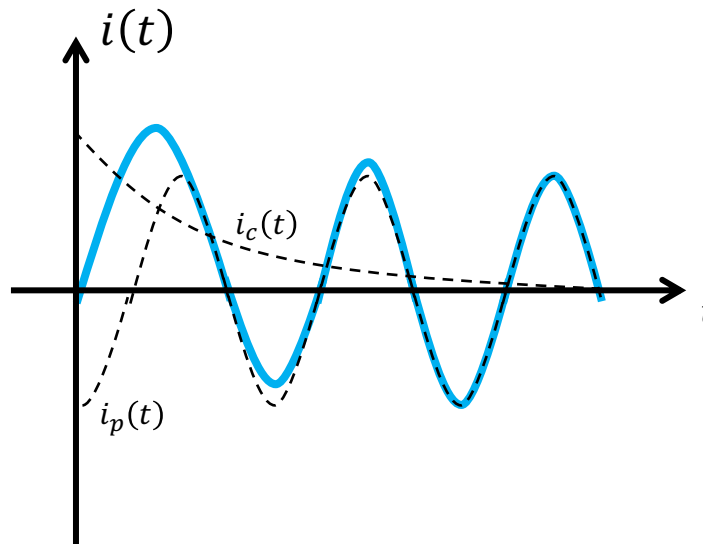
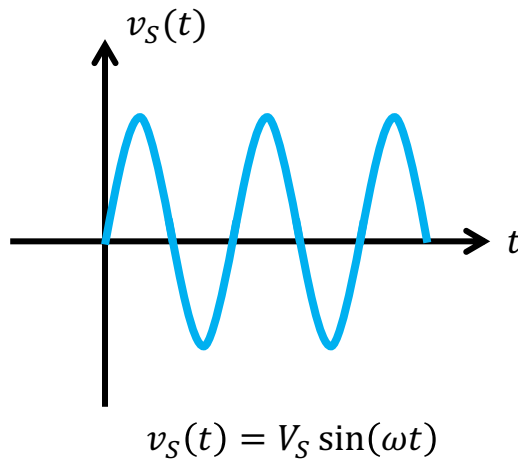
Example 2: Sinusoidal forcing function

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to the sinusoidal forcing function.



$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = \frac{1}{L}v_S(t)$$

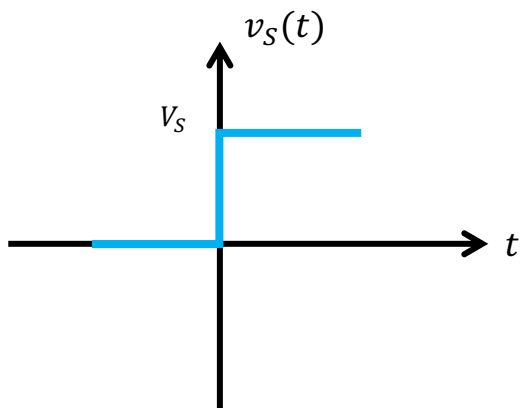
$$\text{Solution } i(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t) + K_3 e^{-\frac{R}{L}t}$$



where

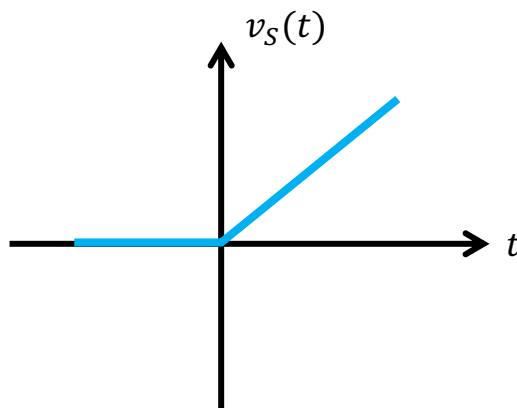
$$\begin{cases} K_1 = V_S \frac{-\omega L}{R^2 + \omega^2 L^2} \\ K_2 = V_S \frac{R}{R^2 + \omega^2 L^2} \\ K_3 = V_S \frac{\omega L}{R^2 + \omega^2 L^2} \end{cases}$$

More forcing functions



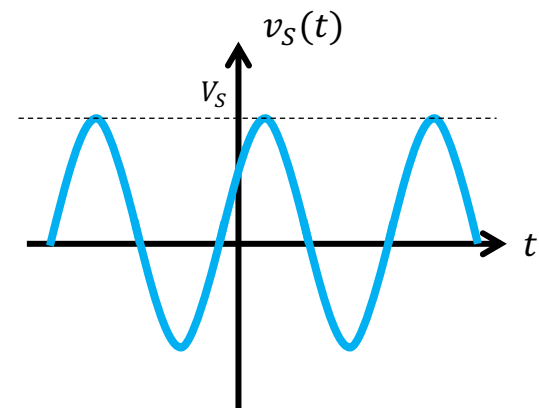
Step function

Assume $x_p(t) = K$



Ramp function

Assume $x_p(t) = pt + q$



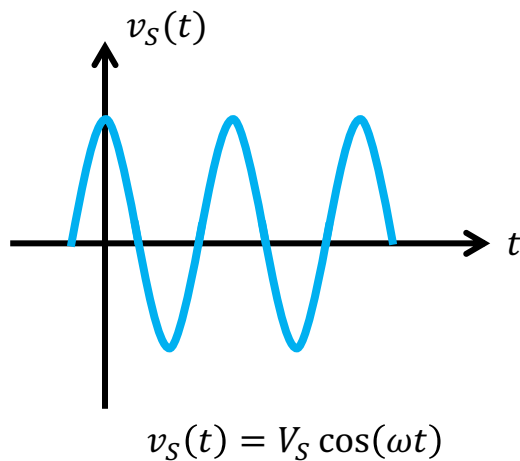
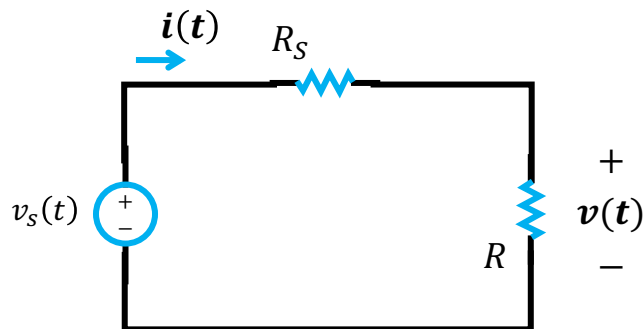
Sinusoidal function

Assume $x_p(t) = K_1 \sin(\omega t) + K_2 \cos(\omega t)$

Outlines

- Response to different forcing function
 - Ramping forcing function
 - Sinusoidal forcing function
- **Complex forcing function**

Resistors with a sin. forcing func.



With $v_{s1}(t) = V_S \cos(\omega t)$

- According to KVL

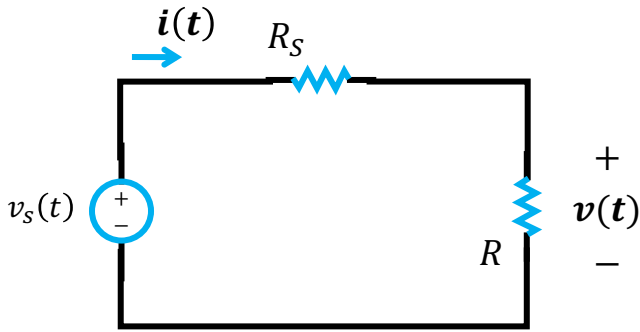
$$v(t) = \frac{R}{R + R_S} v_{s1}(t) = \frac{RV_S}{R + R_S} \cos(\omega t)$$

With $v_{s2}(t) = jV_S \sin(\omega t)$

- According to KVL

$$v(t) = \frac{R}{R + R_S} v_{s2}(t) = j \frac{RV_S}{R + R_S} \sin(\omega t)$$

Resistors with a sin. forcing func.



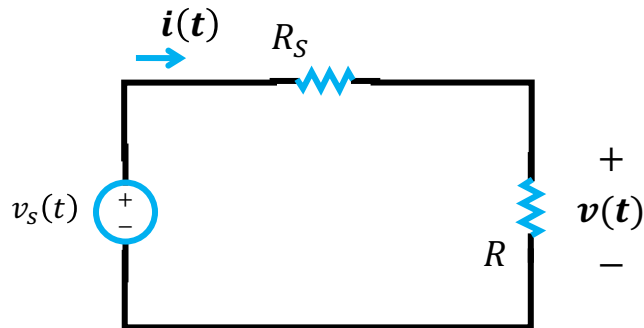
What if $v_s(t) = V_S \cos(\omega t) + jV_S \sin(\omega t)$?

- According to KVL

$$v(t) = \frac{R}{R + R_S} v_s(t)$$

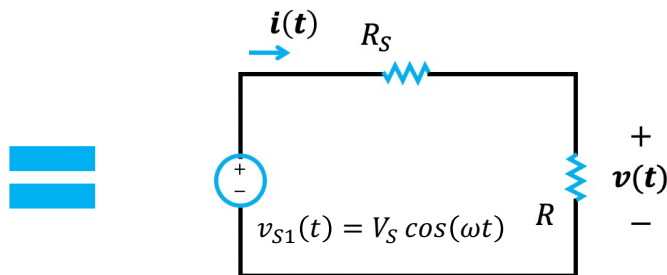
$$= \frac{R}{R + R_S} (V_S \cos(\omega t) + jV_S \sin(\omega t))$$

Resistors with a sin. forcing func.

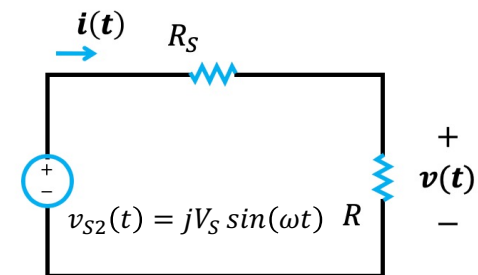


What if $v_s(t) = V_S \cos(\omega t) + jV_S \sin(\omega t)$?

Solution: $v(t) = \frac{R}{R + R_S} (V_S \cos(\omega t) + jV_S \sin(\omega t))$



$$v(t) = \frac{RV_S}{R + R_S} \cos(\omega t)$$



$$v(t) = j \frac{RV_S}{R + R_S} \sin(\omega t)$$

Recall: Euler's eq. & Complex num.

Phasor representation: $A \cos(\omega t + \theta) = |z| \angle \theta$

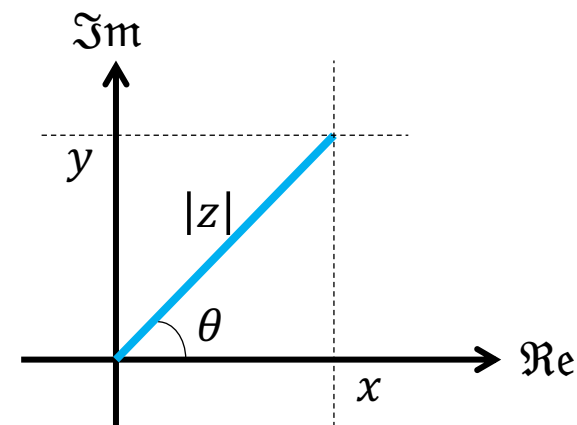
Euler's equation: $e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$

- Real part $\Re[e^{j\omega t}] = \cos(\omega t)$
- imaginary part $\Im[e^{j\omega t}] = \sin(\omega t)$

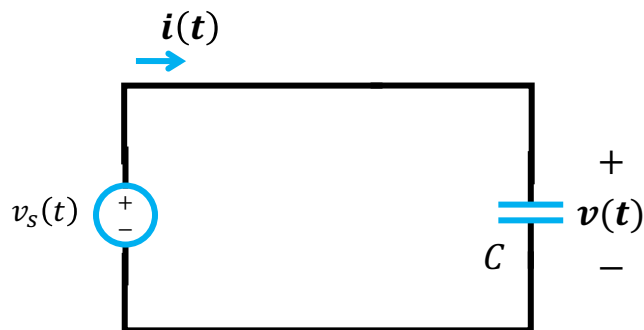
Complex numbers

- Rectangular coordinates $z = x + jy$
- Polar coordinates $z = |z|e^{j\theta}$

Where
$$\begin{cases} |z| = \sqrt{x^2 + y^2} \\ \theta = \tan^{-1}\left(\frac{y}{x}\right) \end{cases}$$



Complex forcing function



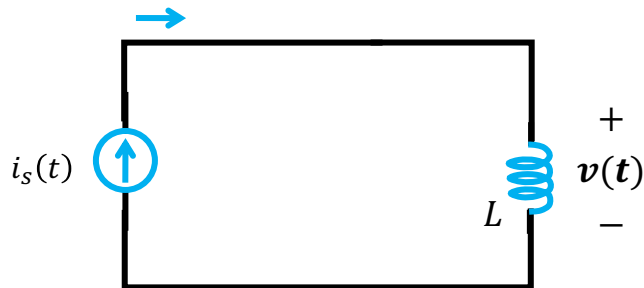
- If a complex function is applied to the capacitor

$$v_s(t) = V_s e^{j\omega t}$$

- According to the $i - v$ relationship of capacitor

$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= C \frac{d}{dt} (V_s e^{j\omega t}) \\ &= j\omega C V_s e^{j\omega t} \end{aligned}$$

Complex forcing function



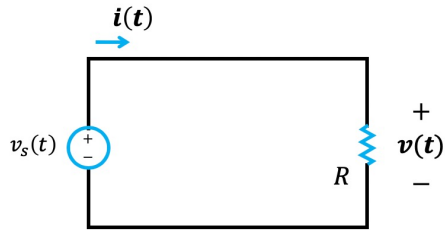
- If a complex function is applied to the inductor

$$i_s(t) = I_s e^{j\omega t}$$

- According to the $i - v$ relationship of inductor

$$\begin{aligned} v(t) &= L \frac{di(t)}{dt} \\ &= L \frac{d}{dt} (I_s e^{j\omega t}) \\ &= j\omega L I_s e^{j\omega t} \end{aligned}$$

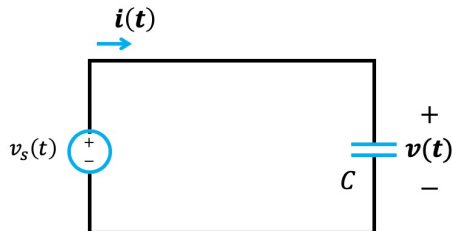
Complex forcing func. on passive dev.



$$v(t) = V_S e^{j\omega t}$$

$$i(t) = \frac{V_S}{R} e^{j\omega t}$$

$$\frac{v(t)}{i(t)} = R$$



$$v(t) = V_S e^{j\omega t}$$

$$i(t) = j\omega C V_S e^{j\omega t}$$

$$\frac{v(t)}{i(t)} = \frac{1}{j\omega C}$$



$$i(t) = I_S e^{j\omega t}$$

$$v(t) = j\omega L I_S e^{j\omega t}$$

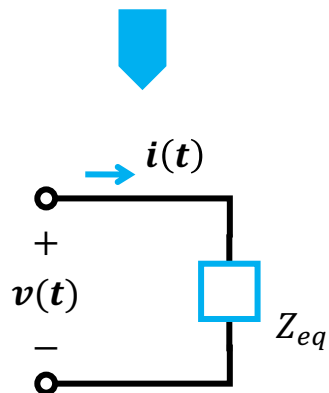
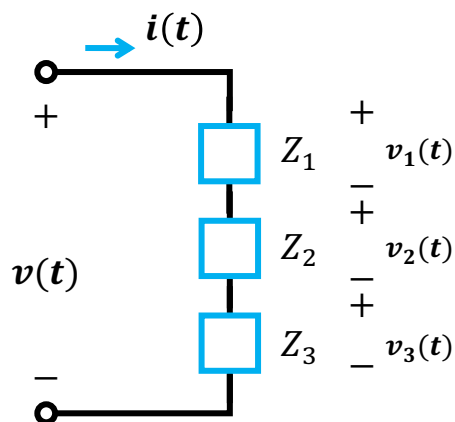
$$\frac{v(t)}{i(t)} = j\omega L$$

Impedance

Impedance, Z ,
is defined as the ratio of the phasor voltage to the phasor current

			
i - v relation	$i = \frac{v}{R}$	$i(t) = C \frac{dv(t)}{dt}$	$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v(x) dx$
v - i relation	$v = iR$	$v(t) = v(t_0) + \frac{1}{C} \int_{t_0}^t i(x) dx$	$v(t) = L \frac{di(t)}{dt}$
Impedance	R	$\frac{1}{j\omega C}$	$j\omega L$

Series connection



According to KVL

$$v(t) = v_1(t) + v_2(t) + v_3(t)$$

According i - v relationship

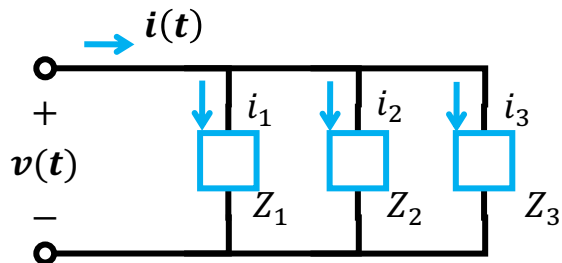
$$\begin{aligned} v(t) &= i(t)Z_1 + i(t)Z_2 + i(t)Z_3 \\ &= i(t)(Z_1 + Z_2 + Z_3) \end{aligned}$$

For the equivalent circuit

$$v(t) = i(t)Z_{eq}$$


$$Z_{eq} = Z_1 + Z_2 + Z_3$$

Parallel connection



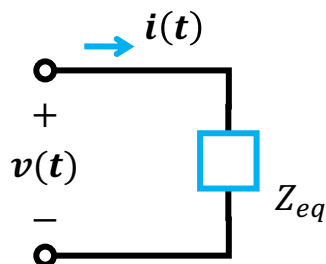
According i - v relationship

$$v(t) = i_1(t)Z_1 = i_2(t)Z_2 = i_3(t)Z_3$$

$$v(t) = i(t)Z_{eq}$$

According to KVL

$$v(t) = Z_{eq}(i_1(t) + i_2(t) + i_3(t))$$

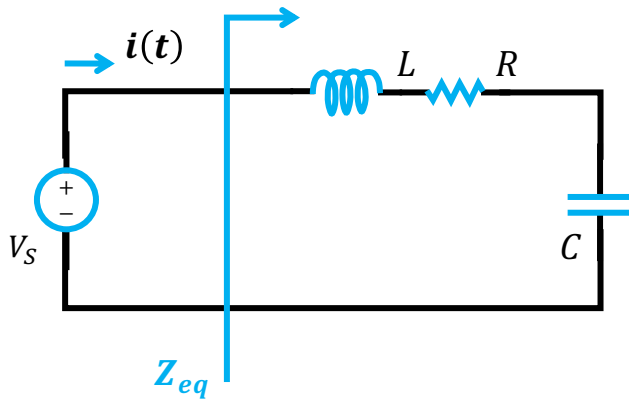


$$= Z_{eq} \left(\frac{v(t)}{Z_1} + \frac{v(t)}{Z_2} + \frac{v(t)}{Z_3} \right)$$

$$\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}$$

Capacitive & Inductive

QUESTION: Find the value of the equivalent impedance, Z_{eq}

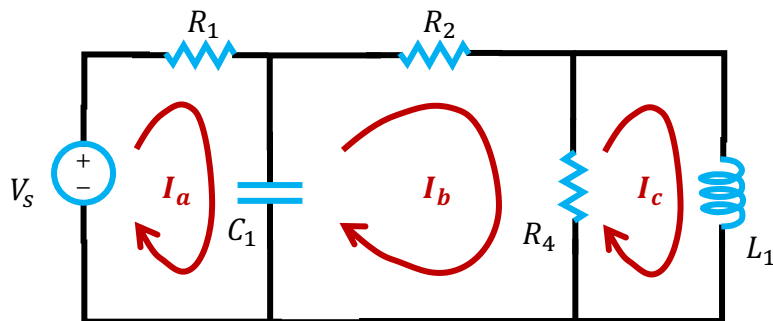


$$Z_{eq} = R + j\omega L + \frac{1}{j\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

- if $\omega L - \frac{1}{\omega C} > 0$ the reactance is inductive
- if $\omega L - \frac{1}{\omega C} < 0$ the reactance is capacitive

Example 4: mesh-current with imp.

QUESTION: Find the output current of the voltage source

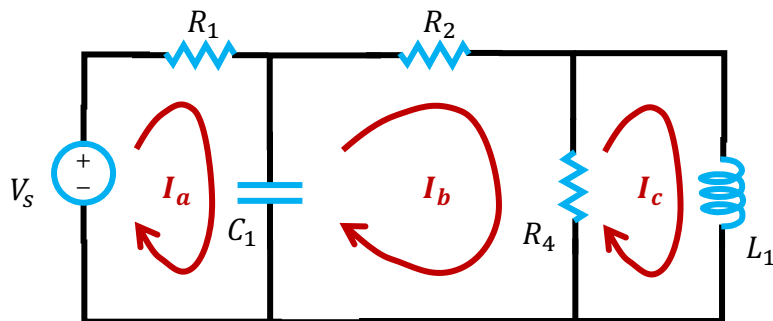


- According to KVL

$$\begin{cases} -V_s + I_a R_1 + (I_a - I_b) \frac{1}{j\omega C} = 0 \\ (I_b - I_a) \frac{1}{j\omega C} + I_b R_2 + (I_b - I_c) R_4 = 0 \\ (I_c - I_b) R_4 + j\omega L I_c = 0 \end{cases}$$

Example 4: mesh-current with imp.

QUESTION: Find the output current of the voltage source

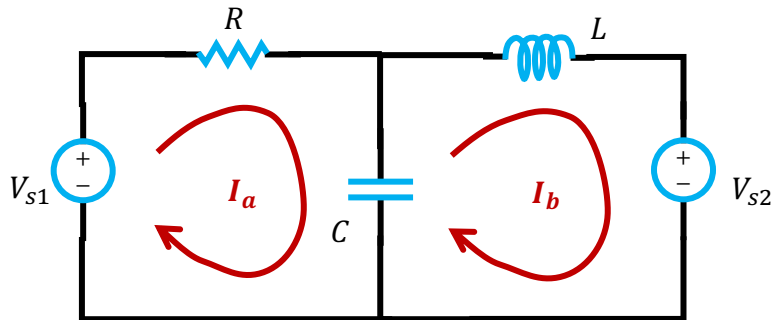


- According to KVL **3 equations in 3 unknowns**

$$\left\{ \begin{array}{l} I_a \left(R_1 + \frac{1}{j\omega C} \right) - I_b \frac{1}{j\omega C} = V_s \\ -I_a \frac{1}{j\omega C} + I_b \left(\frac{1}{j\omega C} + R_2 + R_4 \right) - I_c R_4 = 0 \\ -I_b R_4 + I_c (R_4 + j\omega L) = 0 \end{array} \right.$$

Example 5: Superposition with imp.

QUESTION: Find the voltage on the capacitor C



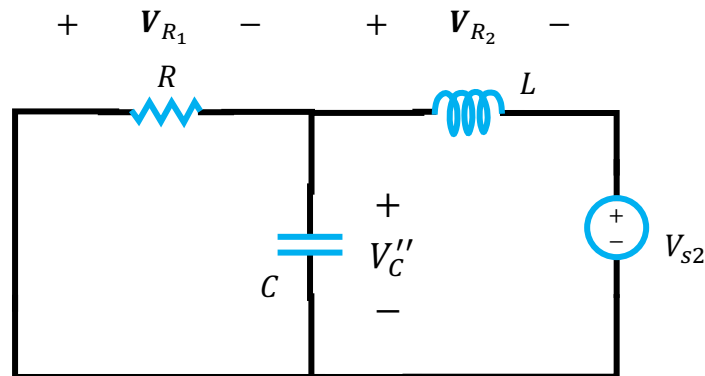
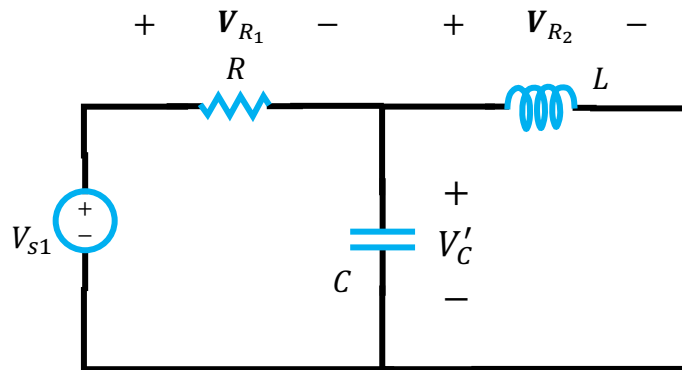
$$\Rightarrow \begin{cases} I_a = \frac{\frac{1}{j\omega C} V_{s1} + \left(R + \frac{1}{j\omega C}\right) V_{s2}}{j\omega LR + \frac{R}{j\omega C} + \frac{L}{C}} \\ I_b = \frac{j\left(\omega L - \frac{1}{\omega C}\right) V_{s1} + \frac{1}{j\omega C} V_{s2}}{j\omega LR + \frac{R}{j\omega C} + \frac{L}{C}} \end{cases}$$

■ According to KVL

$$\begin{cases} -V_{s1} + RI_a + \frac{1}{j\omega C} (I_a - I_b) = 0 \\ \frac{1}{j\omega C} (I_b - I_a) + j\omega LI_b - V_{s2} = 0 \end{cases}$$

$$\Rightarrow V_C = \frac{1}{j\omega C} (I_a - I_b) = \frac{\frac{L}{C} V_{s1} - \frac{R}{j\omega C} V_{s2}}{j\omega LR + \frac{R}{j\omega C} + \frac{L}{C}}$$

Example 5: Superposition with imp.



- According to voltage division

$$V'_C = \frac{j\omega L \parallel \frac{1}{j\omega C}}{R + j\omega L \parallel \frac{1}{j\omega C}} V_{s1}$$

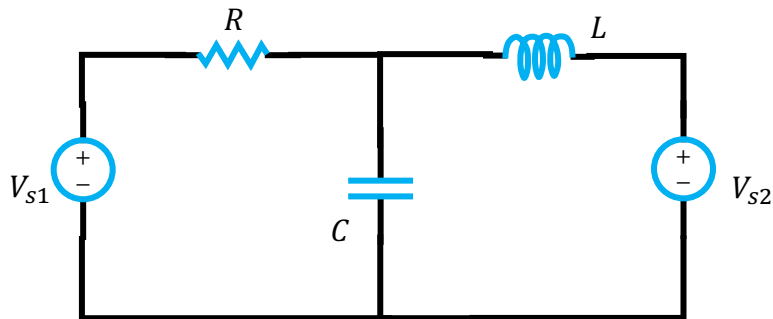
$$= \frac{\frac{L}{C}}{j\omega LR + \frac{R}{j\omega C} + \frac{L}{C}} V_{s1}$$

$$V''_C = -\frac{R \parallel \frac{1}{j\omega C}}{j\omega L + R \parallel \frac{1}{j\omega C}} V_{s2}$$

$$= -\frac{\frac{R}{j\omega C}}{j\omega LR + \frac{R}{j\omega C} + \frac{L}{C}} V_{s2}$$

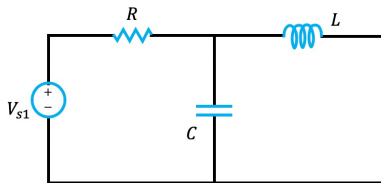
Example 5: Superposition with imp.

QUESTION: Find the voltage on the capacitor C



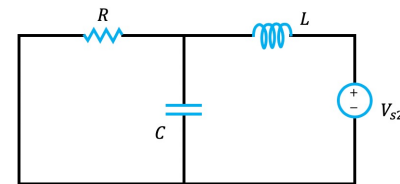
$$V_C = \frac{\frac{L}{C} V_{s1} - \frac{R}{j\omega C} V_{s2}}{j\omega LR + \frac{R}{j\omega C} + \frac{L}{C}}$$

=



$$V'_C = \frac{\frac{L}{C}}{j\omega LR + \frac{R}{j\omega C} + \frac{L}{C}} V_{s1}$$

+

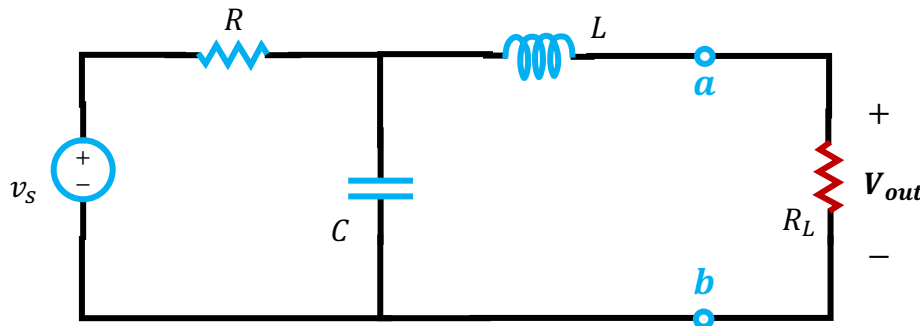


$$V''_C = -\frac{\frac{R}{j\omega C}}{j\omega LR + \frac{R}{j\omega C} + \frac{L}{C}} V_{s2}$$

The superposition property works for ALL LINEAR circuit

Example 6: Thévenin equivalency with imp.

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



Thévenin's theorem

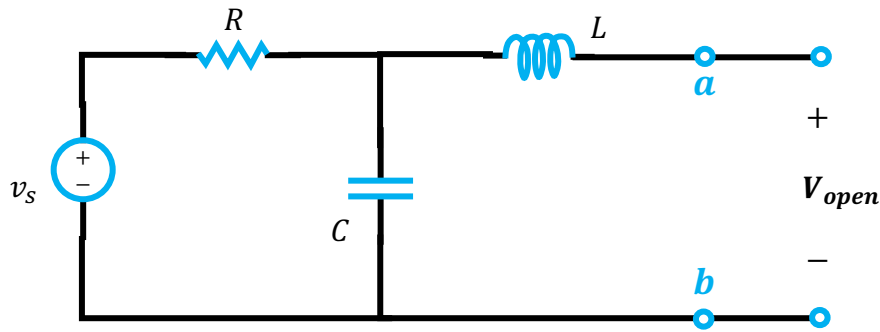
LINEAR two-terminal circuit can be replaced by an equivalent circuit composed of a voltage source and a series impedance

Norton's theorem

LINEAR two-terminal circuit can be replaced by an equivalent circuit composed of a current source and a parallel impedance

Example 6: Thévenin equivalency with imp.

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



- Step 1: remove the load
- Step 2: find V_{open}

- According to voltage division

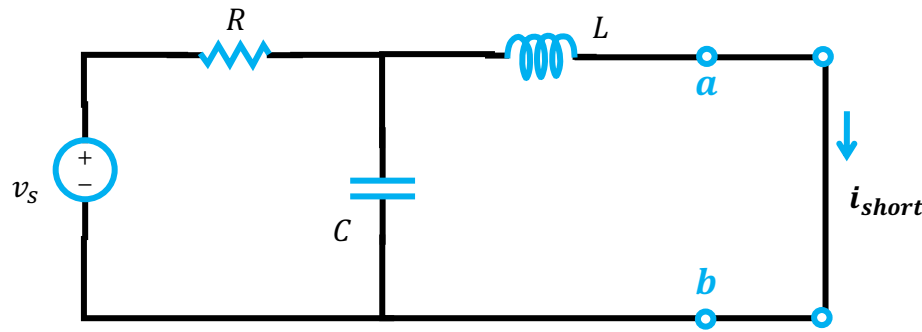
$$V_{open} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} v_s = \frac{1}{1 + j\omega CR} v_s$$



V_{TH}

Example 6: Thévenin equivalency with imp.

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



- Step 1: remove the load
- Step 2: find V_{open}
- Step 3: find i_{short}

- According to Ohm's law

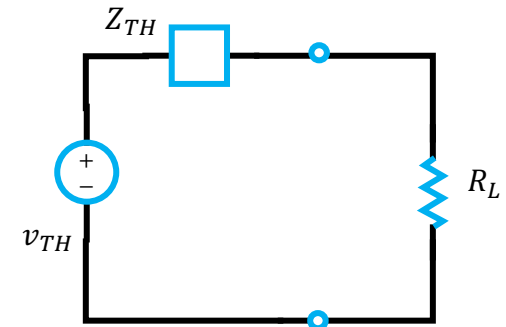
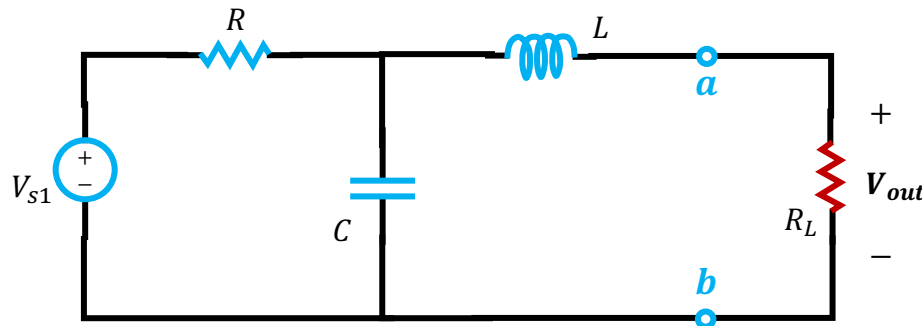
$$i_{open} = \frac{v_s}{R + \frac{1}{j\omega C} || j\omega L} \cdot \frac{\frac{1}{j\omega C}}{\frac{1}{j\omega C} + j\omega L}$$

$$= \frac{v_s}{R + j\omega L - \omega^2 LC}$$

$$i_N$$

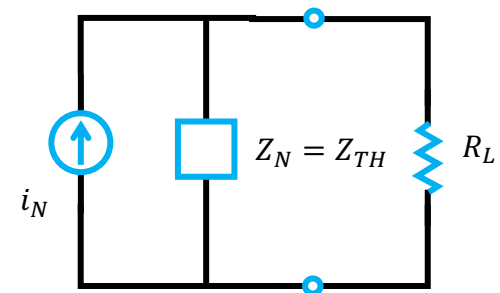
Example 6: Thévenin equivalency with imp.

QUESTION: Find the Thévenin equivalent circuit of the network at the terminals a & b



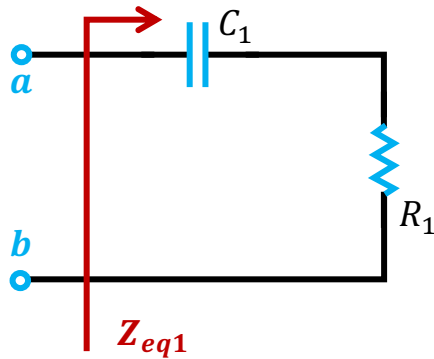
- Step 1: remove the load
- Step 2: find $V_{open} = v_{TH}$
- Step 3: find $i_{short} = i_N$
- Step 4: find Z_{TH}

$$Z_{TH} = \frac{v_{TH}}{i_N} = \frac{R + j\omega L - \omega^2 LC}{1 + j\omega CR}$$



Example 7

QUESTION: Find the value of ω and R_2 , which makes the two circuit equivalent to each other

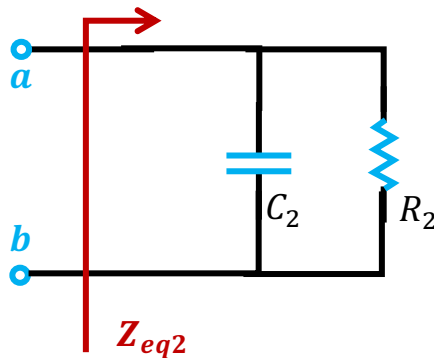


- The equivalent impedance of the upper circuit

$$Z_{eq1} = R_1 + \frac{1}{j\omega C_1}$$

- The equivalent admittance of the lower circuit

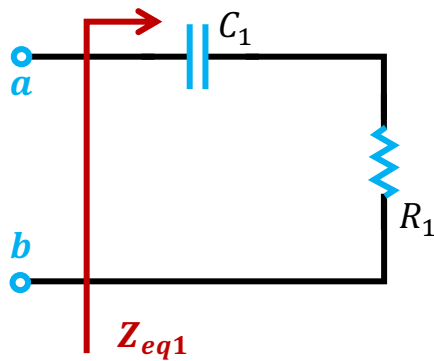
$$G_{eq2} = \frac{1}{R_2} + j\omega C_2$$



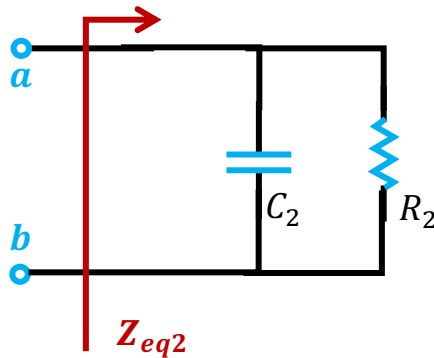
$$Z_{eq1} = \frac{1}{G_{eq2}} \quad \text{when} \quad \begin{cases} \Re[Z_{eq1}] = \Re\left[\frac{1}{G_{eq2}}\right] \\ \Im[Z_{eq1}] = \Im\left[\frac{1}{G_{eq2}}\right] \end{cases}$$

Example 7

QUESTION: Find the value of R_2 and C_2 , which makes the two circuit equivalent to each other



$$G_{eq1} = \frac{1}{R_1 + \frac{1}{j\omega C_1}} = \frac{R_1 + j\frac{1}{\omega C_1}}{R_1^2 + \omega^2 C_1^2}$$
$$= \frac{R_1 \omega^2 C_1^2}{R_1^2 + \omega^2 C_1^2} + j \frac{\omega C_1}{R_1^2 + \omega^2 C_1^2}$$

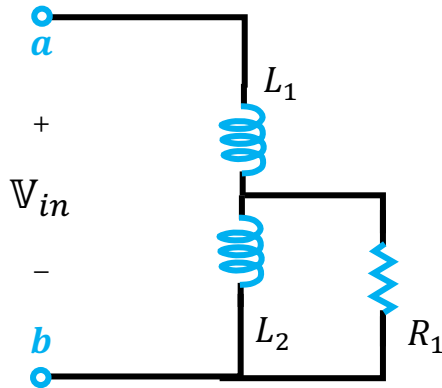


$$G_{eq2} = \frac{1}{R_2} + j\omega C_2$$

$$\Rightarrow \begin{cases} R_2 = \frac{R_1^2 + \omega^2 C_1^2}{R_1 \omega^2 C_1^2} \\ C_2 = \frac{C_1}{R_1^2 + \omega^2 C_1^2} \end{cases}$$

Example 8

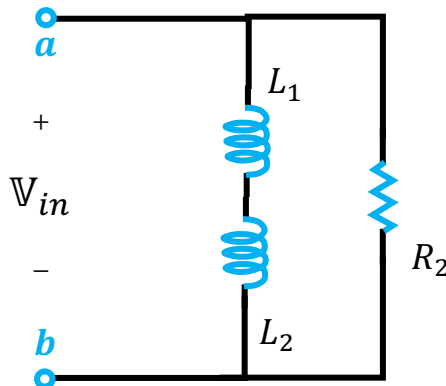
QUESTION: Find the value of R_2 , which makes the power dissipated by R_2 is approximately equivalent to the dissipated by R_1 , when the load resistance R_1 is very high



- According to KVL

$$V_{R_1} = V_{in} \frac{R_1 || Z_{L_2}}{Z_{L_1} + R_1 || Z_{L_2}}$$

$$\xrightarrow{R_1 \gg \omega L_2} V_{in} \frac{Z_{L_2}}{Z_{L_1} + Z_{L_2}} = V_{in} \frac{L_2}{L_1 + L_2}$$



- the power dissipated by R_1 and R_2

$$\begin{cases} P_{R_1} = \frac{V_{R_1}^2}{R_1} \\ P_{R_2} = \frac{V_{in}^2}{R_2} \end{cases}$$

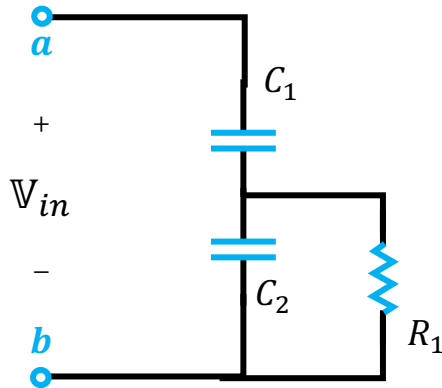


$$R_2 = \frac{R_1}{p^2}$$

$$\text{where } p = \frac{L_2}{L_1 + L_2}$$

Example 9

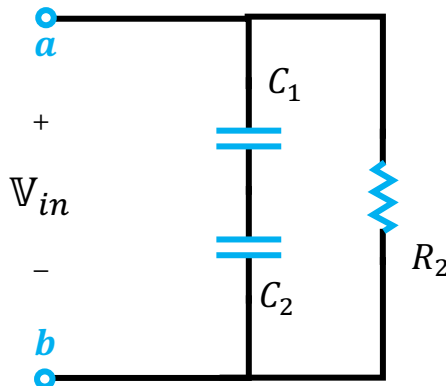
QUESTION: Find the value of R_2 , which makes the power dissipated by R_2 is approximately equivalent to the dissipated by R_1 , when the load resistance R_1 is very high



- According to KVL

$$\mathbb{V}_{R_1} = \mathbb{V}_{in} \frac{R_1 || Z_{C_2}}{Z_{C_1} + R_1 || Z_{C_2}}$$

$$\xrightarrow{R_1 \gg \frac{1}{\omega C_2}} \mathbb{V}_{in} \frac{Z_{C_2}}{Z_{C_1} + Z_{C_2}} = \mathbb{V}_{in} \frac{C_1}{C_1 + C_2}$$



- the power dissipated by R_1 and R_2

$$\begin{cases} P_{R_1} = \frac{\mathbb{V}_{R_1}^2}{R_1} \\ P_{R_2} = \frac{\mathbb{V}_{in}^2}{R_2} \end{cases} \quad \Rightarrow \quad R_2 = \frac{R_1}{p^2}$$

where $p = \frac{C_1}{C_1 + C_2}$

Outlines

- Response to different forcing function

- Ramping forcing function
- Sinusoidal forcing function

- Complex forcing function

- Impedance & admittance
- KVL&KCL / Superposition / Equivalency

			
<i>i-v</i> relation	$i = \frac{v}{R}$	$i(t) = C \frac{dv(t)}{dt}$	$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v(x) dx$
<i>v-i</i> relation	$v = iR$	$v(t) = v(t_0) + \frac{1}{C} \int_{t_0}^t i(x) dx$	$v(t) = L \frac{di(t)}{dt}$
Impedance	R	$\frac{1}{j\omega C}$	$j\omega L$

Outlines

- Response to different forcing function

- Ramping forcing function
- Sinusoidal forcing function

- Complex forcing function

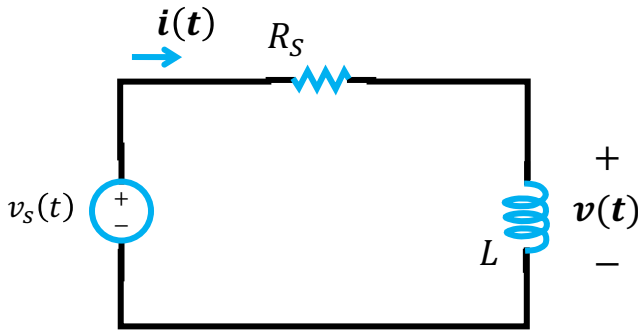
- Impedance & admittance
- KVL&KCL / Superposition / Equivalency

- AC steady-state analysis

			
<i>i-v</i> relation	$i = \frac{v}{R}$	$i(t) = C \frac{dv(t)}{dt}$	$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v(x) dx$
<i>v-i</i> relation	$v = iR$	$v(t) = v(t_0) + \frac{1}{C} \int_{t_0}^t i(x) dx$	$v(t) = L \frac{di(t)}{dt}$
Impedance	R	$\frac{1}{j\omega C}$	$j\omega L$

Example 10

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to a complex forcing function $\mathbb{V}_S = V_S e^{j\omega t}$.



METHODS 1

- According to KVL

$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = \frac{1}{L}v_S(t)$$

- The current through L must be in the form of

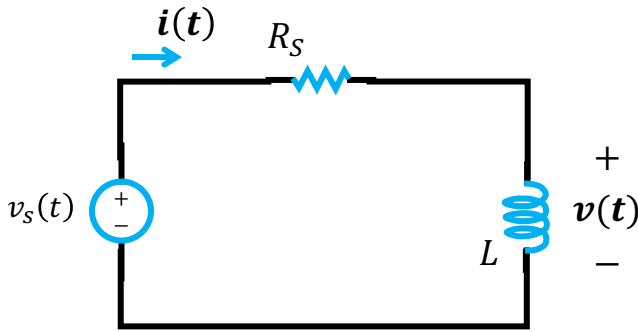
$$i_S(t) = I_S e^{j(\omega t + \varphi)}$$

- Substitute $i_S(t)$ to the equation

$$\frac{d}{dt}(I_S e^{j(\omega t + \varphi)}) + \frac{R}{L}I_S e^{j(\omega t + \varphi)} = \frac{1}{L}V_S e^{j\omega t}$$

Example 10

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to a complex forcing function $\mathbb{V}_S = V_S e^{j\omega t}$.



$$j\omega I_S e^{j(\omega t + \varphi)} + \frac{R}{L} I_S e^{j(\omega t + \varphi)} = \frac{1}{L} V_S e^{j\omega t}$$

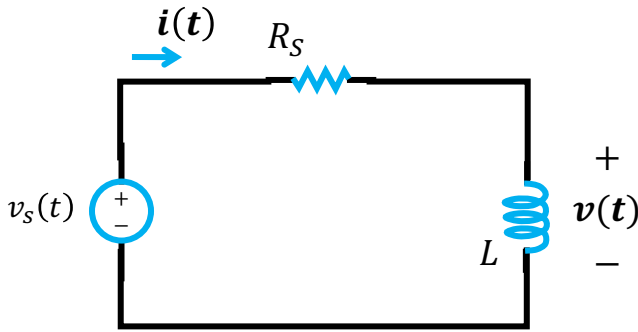
$$j\omega I_S e^{j\varphi} + \frac{R}{L} I_S e^{j\varphi} = \frac{1}{L} V_S$$

$$I_S e^{j\varphi} = \frac{V_S}{R + j\omega L}$$

$$\Rightarrow \begin{cases} I_S = \frac{V_S}{\sqrt{R^2 + \omega^2 L^2}} \\ \varphi = -\tan^{-1} \frac{\omega L}{R} \end{cases}$$

Example 10

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to a complex forcing function $\mathbb{V}_S = V_S e^{j\omega t}$.



$$i_S(t) = I_S e^{j(\omega t + \varphi)}$$

$$= I_S \cos(\omega t + \varphi) + j I_S \sin(\omega t + \varphi)$$

$$\Re[i_S(t)] = I_S \cos(\omega t + \varphi)$$

$$= I_S (\cos \omega t \cos \varphi - \sin \omega t \sin \varphi)$$

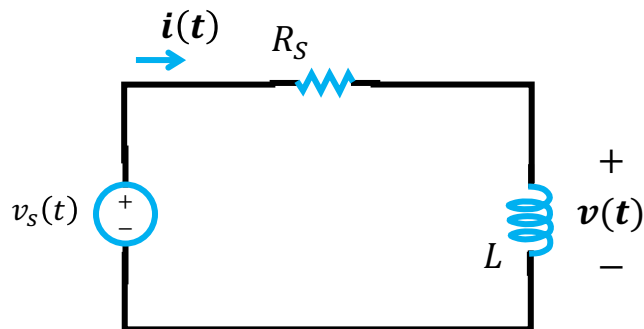
$$= K_1 \cos(\omega t) + K_2 \sin(\omega t)$$

$$\text{where } \begin{cases} K_1 = V_S \frac{R}{R^2 + \omega^2 L^2} \\ K_2 = V_S \frac{\omega L}{R^2 + \omega^2 L^2} \end{cases}$$

$$\begin{cases} I_S = \frac{V_S}{\sqrt{R^2 + \omega^2 L^2}} \\ \varphi = -\tan^{-1} \frac{\omega L}{R} \end{cases}$$

Example 10

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to a complex forcing function $\mathbb{V}_S = V_S e^{j\omega t}$.



METHODS 2

- Consider the impedance circuit
- According to KVL

$$\mathbb{V}_S = R\mathbb{I}_S + j\omega L\mathbb{I}_S \quad \blacktriangleright \quad \mathbb{I}_S = \frac{\mathbb{V}_S}{R + j\omega L}$$

- The real part of $\mathbb{I}_S$

$$\Re[\mathbb{I}_S] = \frac{V_S}{R^2 + \omega^2 L^2} [R \cos(\omega t) - \omega L \sin(\omega t)]$$



$$\Re[i_S(t)]$$

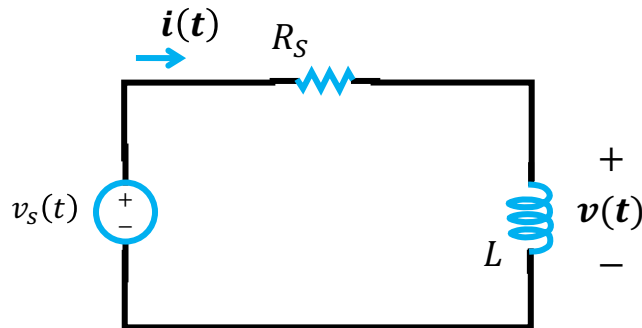
Solution from the differential equation

$$i_S(t) = I_S e^{j(\omega t + \varphi)}$$

$$\left\{ \begin{array}{l} I_S = \frac{V_S}{\sqrt{R^2 + \omega^2 L^2}} \\ \varphi = -\tan^{-1} \frac{\omega L}{R} \end{array} \right.$$

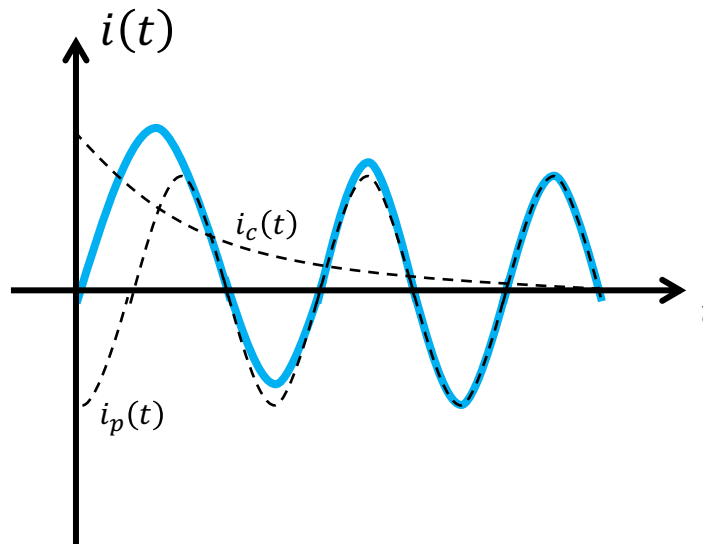
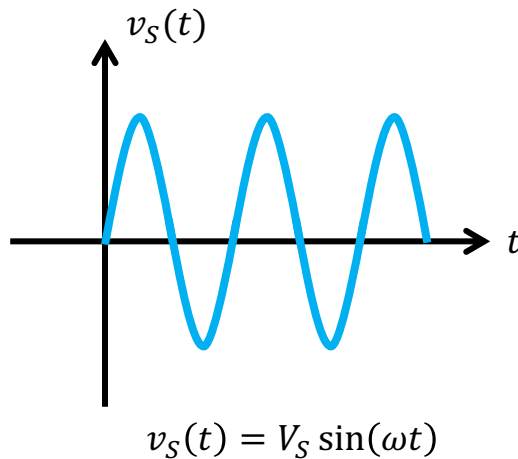
Recall: Sinusoidal forcing function

QUESTION: Assume there is no charge on the inductor L at $t = -\infty$. Find the voltage response to the sinusoidal forcing function.



$$\frac{d}{dt}i(t) + \frac{R}{L}i(t) = \frac{1}{L}v_s(t)$$

Solution $i(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t) + K_3 e^{-\frac{R}{L}t}$

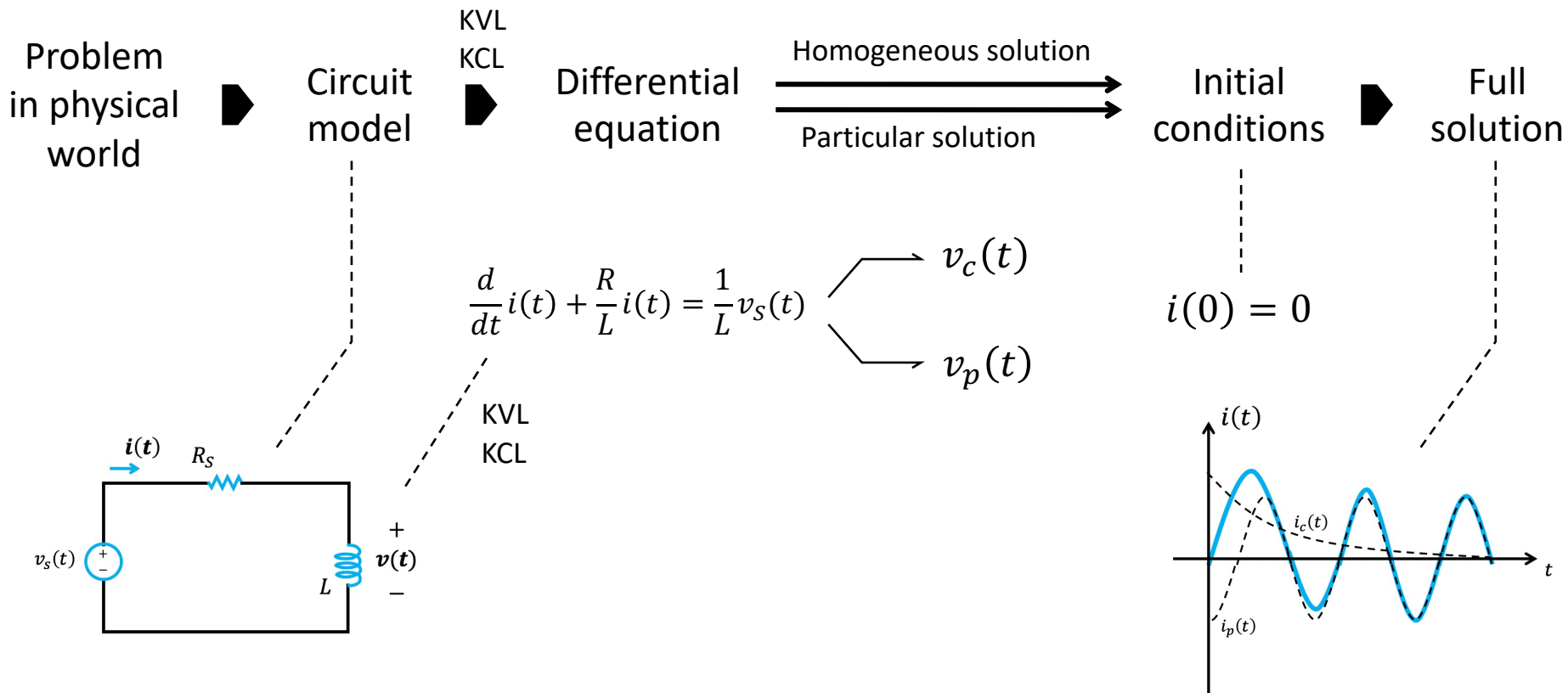


where

$$\begin{cases} K_1 = V_S \frac{-\omega L}{R^2 + \omega^2 L^2} \\ K_2 = V_S \frac{R}{R^2 + \omega^2 L^2} \\ K_3 = V_S \frac{\omega L}{R^2 + \omega^2 L^2} \end{cases}$$

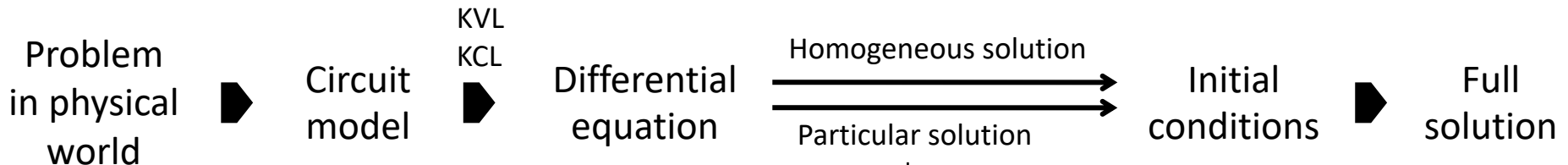
Circuit analysis in TIME domain

Method can be applied to ALL circuit

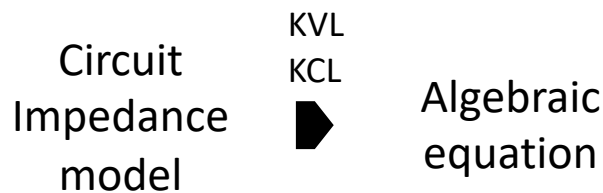


Circuit analysis in TIME domain

Method for ALL circuit



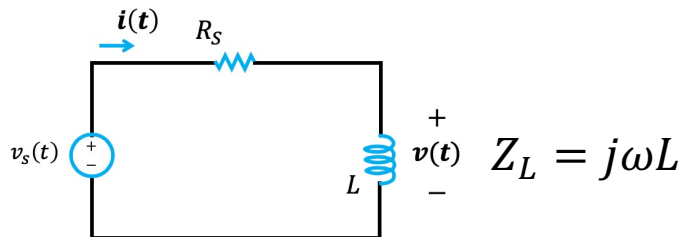
Method ONLY for LINEAR circuit



$\Re[*]$

$$\Re[\mathbb{I}_S] = \frac{V_S}{R^2 + \omega^2 L^2} [R \cos(\omega t) - \omega L \sin(\omega t)]$$

$$\mathbb{V}_S = R\mathbb{I}_S + j\omega L\mathbb{I}_S$$



Outlines

- Response to different forcing function

- Ramping forcing function
- Sinusoidal forcing function

- Complex forcing function

- Impedance & admittance
- KVL&KCL / Superposition / Equivalency

			
<i>i-v</i> relation	$i = \frac{v}{R}$	$i(t) = C \frac{dv(t)}{dt}$	$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v(x) dx$
<i>v-i</i> relation	$v = iR$	$v(t) = v(t_0) + \frac{1}{C} \int_{t_0}^t i(x) dx$	$v(t) = L \frac{di(t)}{dt}$
Impedance	R	$\frac{1}{j\omega C}$	$j\omega L$

- AC steady-state analysis

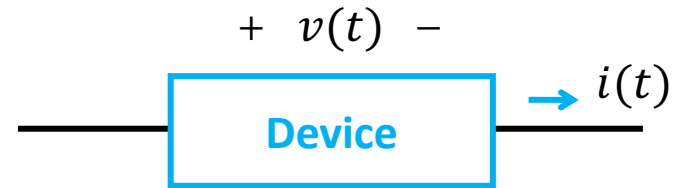
- Impedance circuit analysis for steady-state solution in LINEAR circuit

- Complex Power analysis

Recall: Inst./Avg. power & eff. v/i

Given the voltage $v(t) = V_P \cos(\omega t + \varphi_V)$

Given the current $i(t) = I_P \cos(\omega t + \varphi_I)$



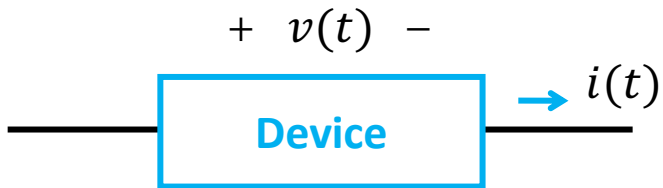
- Instantaneous power
$$p(t) = \frac{1}{2} V_P I_P \cos(\varphi_V - \varphi_I) + \frac{1}{2} V_P I_P \cos(2\omega t + \varphi_V + \varphi_I)$$

- Average power
$$\bar{P} = \frac{V_P}{\sqrt{2}} \frac{I_P}{\sqrt{2}} \cos(\varphi_V - \varphi_I) = v_{rms} i_{rms} \cos(\varphi_V - \varphi_I)$$

- Effective voltage
$$v_{rms} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} v^2(t) dt} = \frac{V_P}{\sqrt{2}}$$

- Effective current
$$i_{rms} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} i^2(t) dt} = \frac{I_P}{\sqrt{2}}$$

Complex power



$$\text{Given } v(t) = V_s \cos(\omega t + \varphi_V) \rightarrow V_s \angle \varphi_V \rightarrow \mathbb{V} = V_s e^{j\varphi_V}$$

$$\text{Given } i(t) = I_s \cos(\omega t + \varphi_I) \rightarrow I_s \angle \varphi_I \rightarrow \mathbb{I} = I_s e^{j\varphi_I}$$

DEFINE COMPLEX POWER $\mathbb{S} = \frac{1}{2} \mathbb{V} \mathbb{I}^*$

$$\mathbb{S} = \frac{1}{2} \mathbb{V} \mathbb{I}^* = \frac{1}{2} (V_s e^{j\varphi_V}) (I_s e^{-j\varphi_I}) = \frac{1}{2} V_s I_s e^{j(\varphi_V - \varphi_I)}$$

$$= \frac{1}{2} V_s I_s [\cos(\varphi_V - \varphi_I) + j \sin(\varphi_V - \varphi_I)]$$

Complex power

$$\mathbb{S} = \underbrace{\frac{1}{2} V_s I_s \cos(\varphi_V - \varphi_I)}_{\text{Average power } \bar{P}} + j \underbrace{\frac{1}{2} V_s I_s \sin(\varphi_V - \varphi_I)}_{\text{Reactive power } Q}$$

$$\text{Average power } \bar{P} \quad + \quad \text{Reactive power } Q \quad = \quad \text{COMPLEX POWER } \mathbb{S}$$

- **Average power** $\bar{P} = \Re\{\mathbb{S}\} = V_{rms} I_{rms} \cos(\varphi_V - \varphi_I) = I_{rms}^2 R \cos(\varphi_V - \varphi_I)$
(unit: watt/W)
- **Reactive power** $Q = \Im\{\mathbb{S}\} = V_{rms} I_{rms} \sin(\varphi_V - \varphi_I) = I_{rms}^2 X \sin(\varphi_V - \varphi_I)$
(unit: Volt-Ampere Reactive/VAR)
- **COMPLEX POWER** $\mathbb{S} = \bar{P} + jQ$
(unit: Volt-Amp/VA)

Complex power

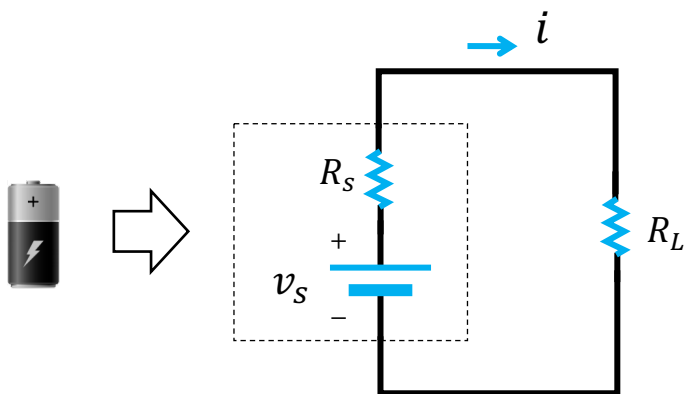
- **COMPLEX POWER** $\mathbb{S} = \bar{P} + jQ$

$$\bar{P} = \Re\{\mathbb{S}\} = V_{rms} I_{rms} \cos(\varphi_V - \varphi_I)$$

$$Q = \Im\{\mathbb{S}\} = j V_{rms} I_{rms} \sin(\varphi_V - \varphi_I)$$

- Define **power factor** $pf = \frac{P}{V_{rms} I_{rms}} = \cos(\varphi_V - \varphi_I)$
- Define **power angle** $\tan(\varphi_V - \varphi_I) = \frac{Q}{P}$

Recall: Max. Power Trans. in DC Circ.



Practical voltage source

- Voltage of R_L

$$v_{R_L} = v_s - iR_s$$

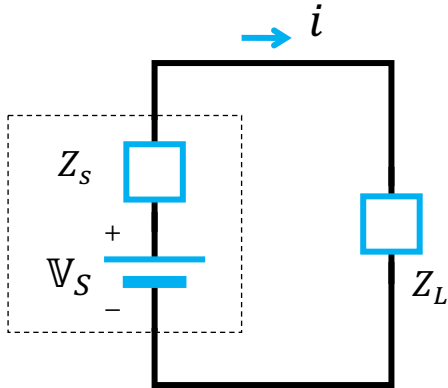
- Power at the load R_L

$$P_L = v_{R_L} i = (v_s - iR_s) i = -R_s \left(i^2 - \frac{v_s}{R_s} i \right) = -R_s \left(i - \frac{1}{2} \frac{v_s}{R_s} \right)^2 + \frac{1}{4} \frac{v_s^2}{R_s}$$
$$\leq \frac{1}{4} \frac{v_{s,rms}^2}{R_s}$$

The maximum power being absorbed by the load

- When $R_s = R_L$ $P_L = P_{L,max} = \frac{1}{4} \frac{v_{s,rms}^2}{R_s}$

Maximum Average Power Transfer



Assume

$$\mathbb{V}_S = V_S \angle \theta_{v_S}$$

$$Z_S = R_S + jX_S$$

$$Z_L = R_L + jX_L$$

- The current through Z_L is $\mathbb{I}_L = I_L \angle \theta_{i_L}$

$$\mathbb{I}_L = \frac{\mathbb{V}_S}{Z_S + Z_L} = \frac{\mathbb{V}_S}{(R_S + jX_S) + (R_L + jX_L)}$$

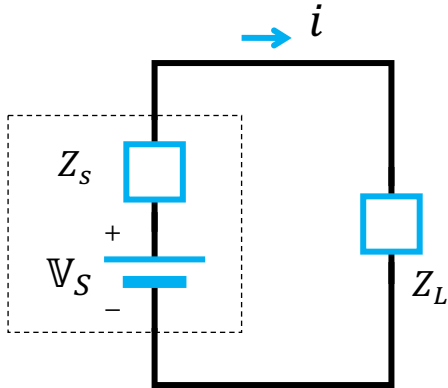
$$\rightarrow I_L = \frac{V_S}{\sqrt{(R_S + R_L)^2 + (X_S + X_L)^2}}$$

- The voltage on Z_L is $\mathbb{V}_L = V_L \angle \theta_{v_L}$

$$\mathbb{V}_L = \frac{\mathbb{V}_S Z_L}{Z_S + Z_L} = \frac{\mathbb{V}_S (R_L + jX_L)}{(R_S + jX_S) + (R_L + jX_L)}$$

$$\rightarrow V_L = \frac{V_S \sqrt{R_L^2 + X_L^2}}{\sqrt{(R_S + R_L)^2 + (X_S + X_L)^2}}$$

Maximum Average Power Transfer



$$\begin{cases} I_L = \frac{V_S}{\sqrt{(R_S + R_L)^2 + (X_S + X_L)^2}} \\ V_L = \frac{V_S \sqrt{R_L^2 + X_L^2}}{\sqrt{(R_S + R_L)^2 + (X_S + X_L)^2}} \end{cases}$$

- The average power at the load Z_L

$$P_L = V_{rms} I_{rms} \cos(\theta_{v_L} - \theta_{i_L}) \quad \text{where} \quad \theta_{v_L} - \theta_{i_L} = \cos^{-1} \left(\frac{R_L}{\sqrt{R_L^2 + X_L^2}} \right)$$

$$= \frac{V_{rms}^2 R_L}{(R_S + R_L)^2 + (X_S + X_L)^2} \leq \frac{V_{rms}^2 R_L}{(R_S + R_L)^2} \leq \frac{1}{4} \frac{V_{rms}^2}{R_S}$$

- when $\begin{cases} R_S = R_L \\ X_S = -X_L \end{cases} \rightarrow \mathbf{Z_S = Z_L^*} \quad P_{L,max} = \frac{1}{4} \frac{V_{rms}^2}{R_S}$

Outlines

- Response to different forcing function

- Ramping forcing function
- Sinusoidal forcing function

- Complex forcing function

- Impedance & admittance
- KVL&KCL / Superposition / Equivalency

- AC steady-state analysis

- Impedance circuit analysis for steady-state solution in LINEAR circuit

- Complex Power analysis

- Definition of complex power/average power/reactive power
- Maximum average power transfer

			
<i>i-v</i> relation	$i = \frac{v}{R}$	$i(t) = C \frac{dv(t)}{dt}$	$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v(x) dx$
<i>v-i</i> relation	$v = iR$	$v(t) = v(t_0) + \frac{1}{C} \int_{t_0}^t i(x) dx$	$v(t) = L \frac{di(t)}{dt}$
Impedance	R	$\frac{1}{j\omega C}$	$j\omega L$

Reading tasks & learning goals

- Reading tasks

- Basic Engineering Circuit Analysis, 10th edition
 - Chapter 8.1-8.8 & 9.1-9.6

- Learning goals

- Know how to solve response with **ramp/sin. forcing functions**
- Understand the basic characteristics of sinusoidal functions
- Know how to calculate **impedance/admittance**
- Know how to perform **AC steady-state analysis**
- Know how to calculate **real power/reactive power/complex power/power factor** in AC circuits
- Be able to calculate the **maximum average power transfer** for a load in an AC circuit